Q2.

$$V_t = \sum_{i=1}^{3} \lambda_t^{(i)} S_t^{(i)} \quad \text{and} \quad w_i = \lambda_t^{(i)} S_t^{(i)} / V_t$$

$\lambda_t^{(i)} S_t^{(i)} \propto cap_i$

$$w_i = \frac{cap_i}{\sum_j cap_j}, \quad \sum cap_j = 3.109 + 4.536 + 4.640 = 12.285$$

$$w_{MSFT} = \frac{3.109}{12.285} = 0.2531, \quad w_{AAPL} = \frac{4.536}{12.285} = 0.3692$$

$$w_{GooGL} = \frac{4.640}{12.285} = 0.3777$$

With $V_t = \$ = 1{,}000{,}000$ at the 05/22/2026 close

$$\lambda_{MSFT} = \frac{253{,}073}{412.57} = 604.6, \quad \lambda_{AAPL} = \frac{369231}{308.82} = 1195.6$$

$$\lambda_{googl} = \frac{377{,}696}{382.97} = 986.2$$

Q3

$Z_t^{(i)} = \ln S_t^{(i)}$, so $V_t = f(Z_t)$ with $f(z) = \sum_i \lambda^{(i)} e^{z_i}$

Log returns: $x_i = \ln\left(S_{t+\Delta}^{(i)} / S_t^{(i)}\right)$ and $S_{t+\Delta}^{(i)} = S_t^{(i)} e^{x_i}$

$$V_{t+\Delta} = \sum \lambda^{(i)} S_t^{(i)} e^{x_i} = V_t \sum_i w_i e^{x_i}, \quad L_{[t]}(x) = V_t - V_{t+\Delta} =$$

$$= -V_t \sum_i w_i (e^{x_i} - 1) \qquad (\textstyle\sum_i w_i = 1)$$

Linearized:

$$f_{z_i}(Z_t) = \lambda^{(i)} e^{z_t^{(i)}} = \lambda^{(i)} S_t^{(i)} S_t^{(i)} = V_t w_i \text{ and } f_t = 0, \text{ so}$$

$$L_{[t]}^{lin}(x) = -\sum_t f_z(Z_t) x_i = -V_t \sum_i w_i x_i = -V_t w^T x.$$

Quadratic

$$f_{z_i z_j}(Z_t) = \delta_{ij} \lambda^{(i)} S_t^{(i)} = \delta_{ij} V_t w_i, \text{ so the second-order}$$

term is $-\frac{1}{2} \sum_{i,j} f_{z_i z_j} x_i x_j = \frac{1}{2} V_t \sum_i w_i x_i^2$ and

$$L_{[t]}^{quad}(x) = -V_t \sum_t w_i \left(x_i + \tfrac{1}{2} x_i^2\right)$$

These are the exact loss and its first- and second order Taylor. Since $e^x - 1 = x + \frac{1}{2}x^2 + O(x^3)$. As $e^x - 1 \geq x$, $L^{lin} \geq$ pointwise, so linearized Var can only overstate

Q5

Assume $X_{t+\Delta} \mid F_t \sim N(\mu, \Sigma)$, linearized loss $L^{lin} = a^T X_{t+\Delta}$ with $a = -V_t w$ is a linear comb. of jointly normal.

$$L^{lin} \sim N(a^T\mu, a^T\Sigma a) = N(-V_t w^T\mu, V_t^2 w^T \Sigma w)$$

For $L \sim N(m, s^2)$, $F_L(l) = \Phi((l-m)/s) \geq \alpha \iff l \geq m + s\Phi^{-1}(\alpha)$, so $VaR_\alpha(L) = m + s\Phi^{-1}(\alpha)$.

$$Var(L^{lin}) = -V_t w^T\mu + V_t \sqrt{w^T\Sigma w}\,\Phi^{-1}(\alpha).$$

Using estimates:

$$VaR_{0.95} = -2,711 + 10^6 \times 0.011904 \times 1.6449 = 16,869$$